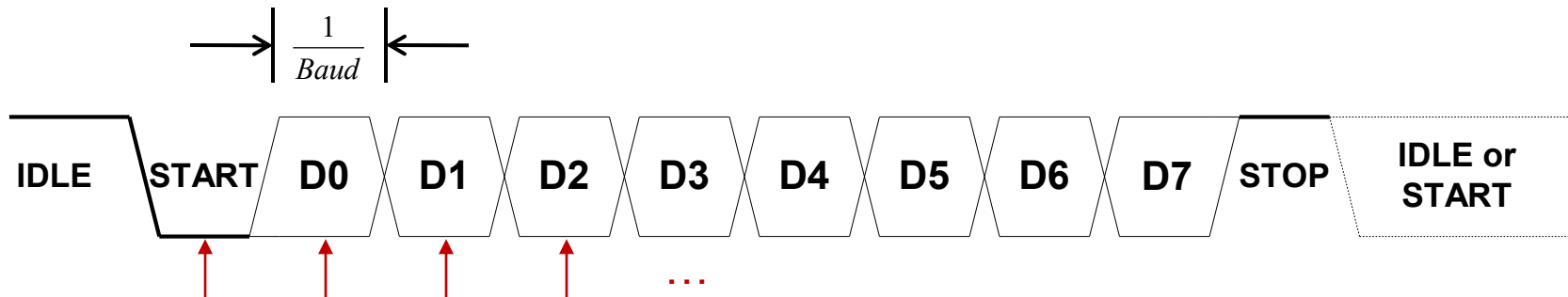
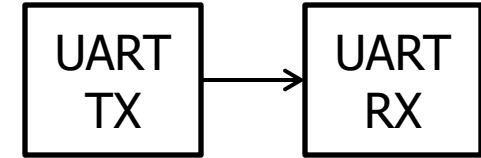




Exercise 12: UART Receiver

Same rate & format as UART_tx:

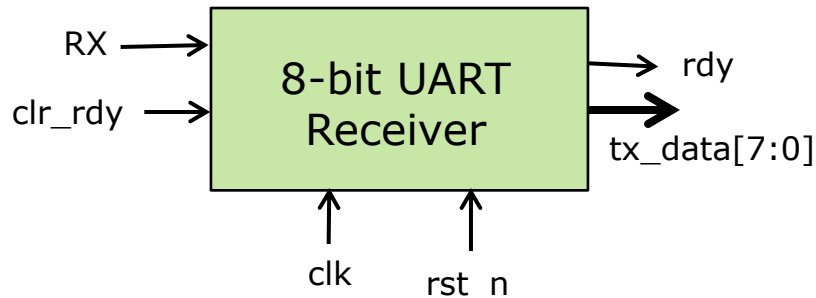
- Baud rate: 19200
- Format: 8N1 (1 start, no parity, 1 stop)
- Receiver
 - Monitors for falling edge of Start bit
 - Sample RX data in the middle of Baud period





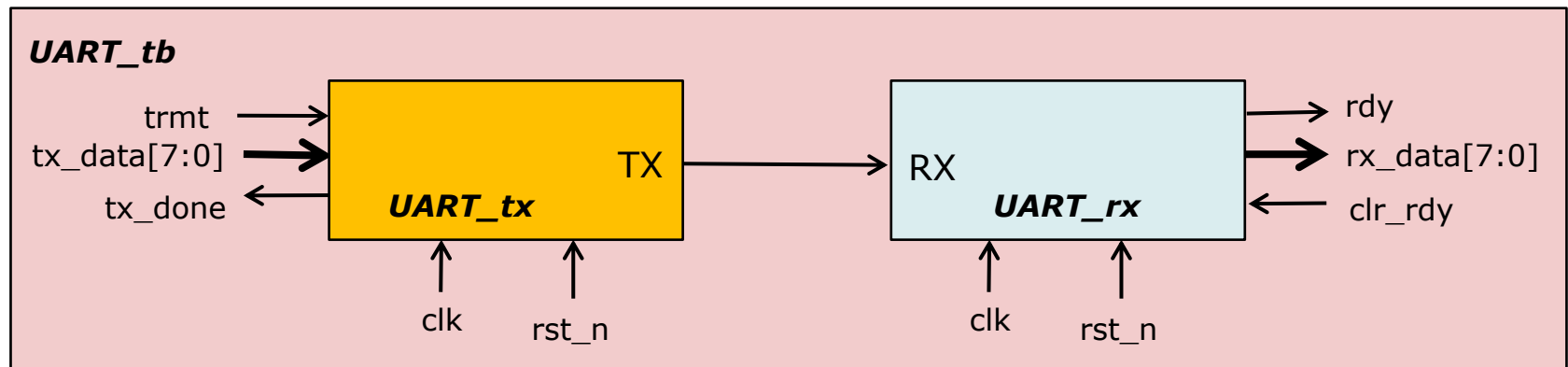
Exercise 12: UART Receiver

- Implement a UART receiver (**UART_rx.sv**) with the following interface:



Signal	Dir	Description
clk,rst_n	in	50MHz system clock & active low reset
RX	in	Serial data input
clr_rdy	In	Knocks down rdy when asserted
rx_data[7:0]	out	Byte received
rdy	out	Asserted when byte received. Stays high till start bit of next byte starts, or until clr_rdy asserted.

- Implement self-checking testbench (**UART_tb.sv**)





Ex12: UART Receiver Design

Where to start?

- We need to take serial data & store it as parallel bits
 - Something to hold data → registers
 - Update register 1 bit at a time → shift register



- Need to keep track of which bit we are at → counter
- Need to keep track of Baud rate → counter
- Need FSM to manage control signals
- Need synchronizer to handle metastability issue for RX



Possible Topology of UART_rx

